

[UNI/ELECENG 3TP3/ELECENG 3TP3.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/ZIR-REAL.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/ZIR-REPEATED.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/ZIR-COMPLEX.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/GRAPHICAL-CONVOLUTION.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/IMPULSE-RESPONSE-LONG.MD](#)

[UNI/ELECENG 3TP3/EXAMPLES/IMPULSE-RESPONSE-SHORT.MD](#)

ELECENG 3TP3

Signal: set of information

System: process that modifies signals or extracts information from them

$x(t)$ is the size of a signal.

Energy E_x of a real-valued signal $x(t)$: $E_x = \int_{-\infty}^{\infty} x^2(t) dt$

Energy E_x of a complex-valued signal $x(t)$: $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$

Average power P_x of a real-valued signal $x(t)$: $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$

Average power P_x of a complex-valued signal $x(t)$: $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$

RMS value of a signal $x(t)$: $x_{\text{RMS}} = \sqrt{P_x}$

For power to exist, the signal must have regularity (eg. periodic).

Analog signals: All values are possible, continuous-time.

Digital signals: Only certain values are possible, continuous-time.

Analog and discrete-time: Any value, $x(t)$ exists only at certain times

Digital and discrete-time: Only certain values, $x(t)$ exists only at certain times

Periodic signals: $x(t) = x(t + T_0)$ for all t . The period is the smallest value of T_0 that satisfies the equation

Casual signals: $x(t) = 0$ for $t < 0$

Non-casual signals: $x(t)$ that starts before $t = 0$

Anti-casual signals: $x(t) = 0$ for $x \geq 0$

Energy signals: Signals with finite energy

Power signals: Signals with finite and non-zero power

An energy signal cannot be a power signal and vice versa, but a signal can also be neither.

Shifting: $\phi(t) = x(t - T)$: shifted version of $x(t)$, $T > 0$ is right shift (time delay), $T < 0$ is left shift (time advance).

Shift and time scaling: $\phi(t) = x(at - b)$ Two options: time shift by b , then time-scale by a . Or time-scale by a , then shift by $\frac{b}{a}$.

The unit impulse $\delta(t)$: $\delta(t) = 0$ for $t \neq 0$, but $\int_{-\infty}^{\infty} \delta(t) dt = 1$

Multiplication of a function by an impulse: $\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$

Sampling property of $\delta(t)$: $\int_{-\infty}^{\infty} \phi(t)\delta(t - T) dt = \phi(T)$, multiplication by a shifted impulse and the integrating and get you a sample of the function at T

Impulse and unit step are not the same, $\frac{du}{dt} = \delta(t)$ and $\int_{-\infty}^{\infty} \delta(t) dt = u(t)$

Even symmetry: $x(t) = x(-t)$, reflected across the vertical line $t = 0$.

Odd symmetry: $x(t) = -x(-t)$, reflected across the vertical line $t = 0$ and horizontal line $x(t) = 0$.

Odd and even components of a signal: $x_e(t) = \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)]$, first brackets are even, second is odd

Time-invariant system: system parameters do not change with time. Shift in T in input = shift by T in output

Casual systems: Output at t_0 depends only on put up to time t_0

Memoryless systems: output at time t_0 depends only on input at time t_0

Finding response $y(t)$ of a system to input $x(t)$: response = output due to initial conditions + output due to input, and total response = zero-input response + zero-state response

Zero-input response is the output when input $x(t) = 0$, zero-state response is the output when initial conditions are zero. The solutions are the roots of the polynomial $Q(\lambda) = 0$.

There are N distinct solutions for $y_0(t)$ when $x(t) = 0$, $c_N e^{\lambda_N t}$. The formula

$Q(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots = 0$ is the characteristic equation. $\lambda_1, \lambda_2, \dots$ are the eigenvalues or characteristic roots of the system. $e^{\lambda_i t}$ are the modes of the system. Zero-input response is a linear combination of the modes.

- Different real roots:
 - Modes are $e^{\lambda_1 t}, e^{\lambda_2 t}, \dots$
 - ZIR is of form $y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots$
- Repeated real roots:
 - Modes are $e^{\lambda t}, t e^{\lambda t}, \dots t^{r-1} e^{\lambda t}$
 - ZIR is of form $y_0(t) = (c_1 + c_2 t + \dots + c_r t^{r-1}) e^{\lambda t}$
- Complex roots:
 - Modes are complex conjugates $\alpha + j\beta, \alpha - j\beta$
 - ZIR is of form $y_0(t) = c e^{\alpha t} \cos(\beta t + \theta)$

- You will need to solve for θ as well

How to find ZIR:

- Find modes
- Substitute into ZIR form
- Differentiate into all the forms you were given initial conditions for
- Substitute initial conditions and solve for constants (and θ if complex modes)
- ZERO-INPUT RESPONSE - REAL, ZERO-INPUT RESPONSE - REPEATED, ZERO-INPUT RESPONSE - COMPLEX

By splitting the input $x(t)$ to a series of small time intervals and rectangular pulses, through superposition, the output $y(t)$ is a sum of the individual responses. As the time interval becomes small, the rectangles become impulses. Then the input can be viewed as a continuous sum of time-shifted impulses

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

.

The impulse response is the response $y(t) = h(t)$ when the input $x(t)$ is made the dirac impulse $\delta(t)$. The impulse happens at $t = 0$ but the system can keep responding past $t = 0$. In general, the form of

$$h(t) = b_0 \delta(t) + (\text{characteric modes})u(t)$$

Finding impulse response:

- Find characteristic modes (response for $t > 0$ and thus input is zero).
- On the RHS, turn $x(t)$ into $\delta(t)$.
- Try to match coefficients of $\delta(t)$ to $h(t)$ from the LHS to RHS. If there are derivatives of the input $\delta(t)$, use the property $h'(t) \supset f(0^+) \delta(t)$.
 - This property comes from the fact that $h(t)$ has a jump at $t = 0$ of size $h(0^+) - h(0^-) = f(0^+) - 0 = f(0^+)$. A jump is the size of the jump multiplied by a Dirac delta, hence the derivative will contain the jump term. Differentiate again and get $h'' \supset f(0^+) \delta'(t)$. There is also a possible jump in $h'(t)$. If $f'(0^+)$ is the right-hand derivative of f , following the above, $h'' \supset f'(0^+) \delta(t)$. Since h'' now contains two terms, $h'' \supset f(0^+) \delta'(t) + f'(0^+) \delta(t)$. Continue this logic for any more differential terms required.
 - In general, $h^{(n)} = f^{(n)}(t)u(t) + \sum_{k=0}^{n-1} f^{(n-1-k)}(0^+) \delta^{(k)}(t)$
 - For example, $h'''' \supset f(0^+) \delta'''' + f'(0^+) \delta''' + f''(0^+) \delta'' + f'''(0^+) \delta'$
 - Notice how the $h(t)$ term does not have any $\delta(t)$ terms inside it.

- Matching coefficients gives the values for $h^N(0^+)$. Since the characteristic modes are the response for $t > 0$, substitute $t = 0^+$ to find an expression for $h(0^+)$. Differentiate the characteristic modes to get more equations for $h^N(0^+)$ as functions of c_n .
- Solve for the constants and you now have $h(t)$. Don't forget the $u(t)$ on the characteristic modes.
- If $\deg P < \deg Q$, there is no $\delta(t)$ term in $h(t)$.
- **IMPULSE RESPONSE (METHOD 1)**

Alternatively, identifying $Q(D)y(t) = P(D)x(t)$:

- The end goal is $h(t) = b_0\delta(t) + [P(D)y_n(t)]u(t)$ where $y_n(t)$ is the linear combination of the characteristic modes.
 - Find characteristic modes.
 - Form y_n as the linear combination of the characteristic modes.
 - Create the initial conditions based on:
 - $y_n(0) = 0$
 - $y'_n(0) = 0$
 - $\dots$
 - $y_n^{(N-1)}(0) = 1$
 - Differentiate y_n however many times needed, substitute, and solve for constants.
 - Multiply by $P(D)$, a function of differential terms D .
 - Calculate the ensuing differentiation/constant multiplication.
 - The result is $h(t)$.
 - If $\deg P < \deg Q$, there is no $\delta(t)$ term and so $h(t) = [P(D)y_n(t)]u(t)$.

◦ **IMPULSE RESPONSE (METHOD 2)**

Going back to approximating the input with rectangles, each rectangle (called a pulse $p(t)$) has amplitude of $x(t)$ and a narrow width of $\Delta\tau$. The signal can thus be reconstructed by shifting the n -th pulse to the n -th interval and scaling by $x(t)$:

$$x(t) \approx \sum_n x(n\Delta\tau)p(t - n\Delta\tau)$$

Multiply this by $\frac{\Delta\tau}{\Delta\tau}$.

$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) \left[\frac{p(t - n\Delta\tau)}{\Delta\tau} \right] \Delta\tau$$

The term $\frac{p(t-n\Delta\tau)}{\Delta\tau}$ approximates to $\delta(t)$ as $\Delta\tau \rightarrow 0$. Since by definition $\delta(t) \rightarrow h(t)$, and the properties of an LTI system mean all the scaling, shifting, and superposition applied to $\delta(t)$ will apply the same way to $h(t)$:

$$\lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau)\delta(t - n\Delta\tau)\Delta\tau \longrightarrow \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau)h(t - n\Delta\tau)\Delta\tau$$

As $\Delta\tau \rightarrow 0$, $n\Delta\tau$ turns into the continuous τ , $\Delta\tau$ becomes $d\tau$, and the summation becomes a continuous integral sum.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

This is otherwise known as convolution, $y(t) = x(t) * h(t)$.

If you know a LTI system's impulse response $h(t)$, you can find its zero state output $y(t)$ for any $x(t)$ through convolution.

Convolution is:

- Commutative: $x_1(t) * x_2(t) = x_2(t) * x_1(t)$
- Distributive: $x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$
- Associative: $x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$
- Shiftable: if $x_1(t) * x_2(t) = y(t)$:
 - $x_1(t) * x_2(t - T) = x_1(t - T) * x_2(t) = y(t - T)$
 - or $x_1(t - T_1) * x_2(t - T_2) = y(t - T_1 - T_2)$

Convolution with an impulse: $x(t) * \delta(t) = \int x(\tau)\delta(t - \tau) d\tau$

Width property: The width of $x_1(t) * x_2(t)$ is the sum of the two finite widths $T_1 + T_2$ of the two convolved functions.

For two casual signals:

$$\begin{aligned} y(t) = x(t) * h(t) &= \left[\int_0^t x(\tau)h(t - \tau) d\tau \right] u(t) \\ &= \left[\int_0^t h(\tau)x(t - \tau) d\tau \right] u(t) \end{aligned}$$

Graphical method of convolution:

- This method is easiest when the signals are easy to sketch.
- Recall $c(t) = \int_{-\infty}^{\infty} x(\tau)g(t - \tau) d\tau$.
- Given the convolution we are trying to solve has the form $x(t) * g(t) = c(t)$:
 - Substitute $t = \tau$ into $x(t)$ for $x(\tau)$
 - Substitute $t = \tau$ then flip $g(t)$ across y-axis for $g(-\tau)$. Mark $g(-\tau)$'s left and right edges.
 - Shift by an amount t . The edges of $g(-\tau)$ will become $\text{offset}_1 + t < g(\tau) < \text{offset}_2 + t$.
 - A 'region' begins whenever an edge of the moving g signal hits an edge of the stationary x signal and ends right before another edge hit occurs. Break the entire convolution up into regions of t .

- Inside each region, the limits of integration become
 $\tau_{\text{lower}} = \max(\text{left edge of } x, \text{left edge of moving } g),$
 $\tau_{\text{upper}} = \min(\text{right edge of } x, \text{right edge of moving } g).$
- Evaluate $\int_{\tau_{\text{lower}}}^{\tau_{\text{upper}}} x(\tau)g(t-\tau) d\tau$ symbolically to get the formula for that interval.
- If the signals are simple enough, eg. pulses or ramps, you can just vibrate it out by solving for $c(t)$ at boundary t values and connecting the points with straight lines.
- GRAPHICAL CONVOLUTION

Output of a system for exponential input:

$$\begin{aligned} y(t) &= h(t) * e^{st} \\ &= \int h(\tau)e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau)e^{-s\tau} d\tau \\ &= H(s)e^{st} \quad \left(H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau} d\tau \right) \end{aligned}$$

$$\therefore H(s) = \frac{y(t)}{e^{st}} = \frac{\text{output}}{\text{input}}$$

And so $H(s)$ is the **transfer function**.

BIBO stability: bounded input and bounded output. $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$. BIBO stable systems' impulse responses are absolutely integrable.

Asymptotic stability: as $t \rightarrow \infty$, all natural modes $e^{\lambda_i t} \rightarrow 0$.

- Stable: If all $\text{Re}(\lambda_i) < 0$, the mode will decay to zero as $t \rightarrow \infty$.
- Unstable: If at least one $\text{Re}(\lambda_i) > 0$, the mode will grow as $t \rightarrow \infty$ and such the mode is unstable.
 - The system will also be unstable if the roots are repeated and imaginary, since repeated imaginary roots can cause terms like $te^{\lambda t}$ which grow instead of oscillating.
- Asymptotically unstable: Both roots are imaginary and different (will oscillate).

missing rms
missing symmetry
proofread beginning part

Zero-Input Response - Real

Find the zero-input response $y_0(t)$ for the system

$$y''(t) + 5y'(t) + 6y(t) = 0, \quad t \geq 0$$

Given the initial conditions:

- $y(0) = 1$
- $y'(0) = 0$

Solution

1. Set up the ZIR equation

For the zero-input response, $Q(D)y_0(t) = 0$, so we solve the homogeneous differential equation:

$$\begin{aligned} D'' + 5D' + 6D &= 0 \\ \lambda'' + 5\lambda' + 6\lambda &= 0 \end{aligned}$$

Solving the polynomial:

$$\begin{aligned} (\lambda + 2)(\lambda + 3) &= 0 \\ \therefore \lambda_1 = -2, \lambda_2 = -3 \end{aligned}$$

The ZIR form for two real roots is $y_0 = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$. Substituting:

$$y_0(t) = C_1 e^{-2t} + C_2 e^{-3t}$$

2. Apply initial conditions

Using $y(0) = 1$:

$$C_1 + C_2 = 1$$

Differentiate the ZIR form:

$$y'_0(t) = -2C_1 e^{-2t} - 3C_2 e^{-3t}$$

Using $y'(0) = 0$:

$$-2C_1 - 3C_2 = 0$$

Solve the two equations:

$$\begin{cases} C_1 + C_2 = 1 \\ -2C_1 - 3C_2 = 0 \end{cases}$$

Thus,

$$C_2 = -2, \quad C_1 = 3$$

End

Substituting the constants into the general solution gives

$$y_0(t) = 3e^{-2t} - 2e^{-3t}, \quad t \geq 0$$

Zero-Input Response - Repeated

Find the zero-input response $y_0(t)$ for the system

$$y'''(t) + 3y''(t) + 3y'(t) + y(t) = 0, \quad t \geq 0$$

Given the initial conditions:

- $y(0) = 1$
- $y'(0) = 0$
- $y''(0) = 1$

Solution

1. Set up the ZIR equation

For the zero-input response, $Q(D)y_0(t) = 0$, so we solve the homogeneous differential equation:

$$\begin{aligned} D^3 + 3D^2 + 3D + 1 &= 0 \\ \lambda^3 + 3\lambda^2 + 3\lambda + 1 &= 0 \end{aligned}$$

Factoring the characteristic polynomial:

$$\begin{aligned} (\lambda + 1)^3 &= 0 \\ \therefore \lambda &= -1 \quad (\text{with multiplicity } 3) \end{aligned}$$

The ZIR form for a repeated root of multiplicity three is $y_0(t) = (C_1 + C_2t + C_3t^2)e^{\lambda t}$. Substituting $\lambda = -1$:

$$y_0(t) = (C_1 + C_2t + C_3t^2)e^{-t}$$

2. Apply initial conditions

Using $y(0) = 1$:

$$C_1 = 1$$

Differentiate the ZIR form:

$$y'_0(t) = [C_2 + 2C_3t - (C_1 + C_2t + C_3t^2)]e^{-t}$$

Using $y'(0) = 0$:

$$C_2 - C_1 = 0$$

Differentiate again:

$$y''_0(t) = [2C_3 - 2(C_2 + 2C_3t) + (C_1 + C_2t + C_3t^2)]e^{-t}$$

Using $y''(0) = 1$:

$$2C_3 - 2C_2 + C_1 = 1$$

Solve the three equations:

$$\begin{cases} C_1 = 1 \\ C_2 - C_1 = 0 \\ 2C_3 - 2C_2 + C_1 = 1 \end{cases}$$

Thus,

$$C_1 = 1, \quad C_2 = 1, \quad C_3 = 1$$

End

Substituting the constants into the general solution gives

$$y_0(t) = (1 + t + t^2)e^{-t}, \quad t \geq 0$$

Zero-Input Response - Complex

Find the zero-input response $y_0(t)$ for the system

$$y''(t) + 4y'(t) + 13y(t) = 0, \quad t \geq 0$$

Given the initial conditions:

- $y(0) = 1$
- $y'(0) = 1$

Solution

1. Set up the ZIR equation

For the zero-input response, $Q(D)y_0(t) = 0$, so we solve the homogeneous differential equation:

$$\begin{aligned} D^2 + 4D + 13 &= 0 \\ \lambda^2 + 4\lambda + 13 &= 0 \end{aligned}$$

Solving the characteristic equation:

$$\begin{aligned} \lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} \\ &= \frac{-4 \pm \sqrt{-36}}{2} \\ &= -2 \pm 3i \end{aligned}$$

The ZIR form for complex roots $\lambda = \alpha \pm \beta i$ is $y_0(t) = e^{\alpha t}(C_1 \cos(\beta t) + C_2 \sin(\beta t))$. Substituting $\alpha = -2$ and $\beta = 3$:

$$y_0(t) = e^{-2t}(C_1 \cos(3t) + C_2 \sin(3t))$$

2. Apply initial conditions

Using $y(0) = 1$:

$$C_1 = 1$$

Differentiate the ZIR form:

$$y'_0(t) = e^{-2t}[-2(C_1 \cos(3t) + C_2 \sin(3t)) - 3C_1 \sin(3t) + 3C_2 \cos(3t)]$$

Using $y'(0) = 1$:

$$-2C_1 + 3C_2 = 1$$

Solve the two equations:

$$\begin{cases} C_1 = 1 \\ -2C_1 + 3C_2 = 1 \end{cases}$$

Thus,

$$C_1 = 1, \quad C_2 = 1$$

End

Substituting the constants into the general solution gives

$$y_0(t) = e^{-2t}(\cos(3t) + \sin(3t)), \quad t \geq 0$$

Graphical Convolution

Find $c(t) = x(t) * g(t)$

Given:

- $x(t) = t$ for $0 \leq t \leq 2$
- $g(t) = 1$ for $0 \leq t \leq 1$

Solution

1. Set up the boundaries

The signal $x(\tau) = \tau$ is defined on the fixed interval $[0, 2]$.

The original pulse $g(\tau)$ is nonzero for $0 \leq \tau \leq 1$. After flipping, $g(-\tau)$ is nonzero for $-1 \leq \tau \leq 0$. After shifting, $g(t - \tau)$ is nonzero for $t - 1 \leq \tau \leq t$.

- Trailing edge: $\tau = t - 1$
- Leading edge: $\tau = t$
- Inside the interval: $g(t - \tau) = 1$

2. Determine the critical transition points for t

Set the moving edges equal to the stationary boundaries, $\tau = 0$ and $\tau = 2$:

- Leading edge enters the stationary window: $t = 0$
- Trailing edge enters the stationary window: $t - 1 = 0 \implies t = 1$
- Leading edge leaves the stationary window: $t = 2$
- Trailing edge leaves the stationary window: $t - 1 = 2 \implies t = 3$

The timeline divides into five regions:

- $t < 0$
- $0 \leq t < 1$
- $1 \leq t < 2$
- $2 \leq t < 3$
- $t \geq 3$

3. Compute the integral for each region

Region 1: $t < 0$: No overlap

$$c(t) = 0$$

Region 2: $0 \leq t < 1$: Pulse enters the ramp

The overlap is $\tau \in [0, t]$, and the integrand is

$$x(\tau)g(t - \tau) = \tau \cdot 1 = \tau$$

Therefore,

$$c(t) = \int_0^t \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_0^t = \frac{1}{2} t^2$$

Region 3: $1 \leq t < 2$: Pulse is completely inside the ramp

The overlap is $\tau \in [t-1, t]$. The integrand is τ , so

$$c(t) = \int_{t-1}^t \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_{t-1}^t = \frac{1}{2} (t^2 - (t-1)^2) = t - \frac{1}{2}$$

Notice how the result is linear because a window of fixed width is sliding under a straight ramp.

Region 4: $2 \leq t < 3$: Pulse exits the ramp

The overlap is $\tau \in [t-1, 2]$. Thus,

$$c(t) = \int_{t-1}^2 \tau d\tau = \left[\frac{1}{2} \tau^2 \right]_{t-1}^2 = \frac{1}{2} (4 - (t-1)^2) = -\frac{1}{2} t^2 + t + \frac{3}{2}$$

Region 5: $t \geq 3$: Complete exit

Finally,

$$c(t) = 0$$

End

Therefore the result, when pieced back together, is

$$c(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{2} t^2, & 0 \leq t < 1, \\ t - \frac{1}{2}, & 1 \leq t < 2, \\ -\frac{1}{2} t^2 + t + \frac{3}{2}, & 2 \leq t < 3, \\ 0, & t \geq 3 \end{cases}$$

You can do a check and verify that this solution is perfectly continuous on t .

Impulse Response (Method 1)

Find the impulse response of the zero-initial-condition system

$$y'''(t) + 6y''(t) + 11y'(t) + 6y(t) = x'(t) + 2x(t)$$

For an impulse input, $x(t) = \delta(t)$, the output is $h(t)$ and

$$h'''(t) + 6h''(t) + 11h'(t) + 6h(t) = \delta'(t) + 2\delta(t)$$

Solution

1. Find the characteristic modes

For $t > 0$, the impulse forcing is zero, so we want to solve the homogeneous equation derived from the LHS:

$$\begin{aligned} r^3 + 6r^2 + 11r + 6 &= 0 \\ (r+1)(r+2)(r+3) &= 0 \end{aligned}$$

Therefore, for $t > 0$,

$$h(t) = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{-3t}$$

For the causal impulse response, write this as

$$h(t) = f(t)u(t), \quad f(t) = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{-3t}$$

2. Match the impulse terms

For $h(t) = f(t)u(t)$, recall the terms of its derivatives:

$$\begin{aligned} h' &\supset f(0^+) \delta, \\ h'' &\supset f(0^+) \delta' + f'(0^+) \delta, \\ h''' &\supset f(0^+) \delta'' + f'(0^+) \delta' + f''(0^+) \delta \end{aligned}$$

Therefore, the singular terms on the left-hand side are

$$\begin{aligned} \delta'' &: f(0^+), \\ \delta' &: f'(0^+) + 6f(0^+), \\ \delta &: f''(0^+) + 6f'(0^+) + 11f(0^+) \end{aligned}$$

The right-hand side is $\delta' + 2\delta$, so matching coefficients gives

$$\begin{cases} f(0^+) = 0, \\ f'(0^+) + 6f(0^+) = 1, \\ f''(0^+) + 6f'(0^+) + 11f(0^+) = 2 \end{cases}$$

Hence:

$$f(0^+) = 0, \quad f'(0^+) = 1, \quad f''(0^+) = -4$$

3. Solve

Using the characteristic-mode expression, its derivative, and second derivative then substituting 0^+ ,

$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ -c_1 - 2c_2 - 3c_3 = 1 \\ c_1 + 4c_2 + 9c_3 = -4 \end{cases}$$

which gives

$$c_1 = \frac{1}{2}, \quad c_2 = 0, \quad c_3 = -\frac{1}{2}$$

End

Thus the impulse response is

$$h(t) = \frac{1}{2}(e^{-t} - e^{-3t})u(t)$$

Impulse Response (Method 2)

Find the impulse response of the zero-initial-condition system

$$y'''(t) + 6y''(t) + 11y'(t) + 6y(t) = x'(t) + 2x(t)$$

For an impulse input, $x(t) = \delta(t)$, the output is $h(t)$ and

$$h'''(t) + 6h''(t) + 11h'(t) + 6h(t) = \delta'(t) + 2\delta(t)$$

For an impulse input, the output is $h(t)$.

Solution

1. Find the characteristic modes

The characteristic equation for $Q(D)$ is

$$\begin{aligned} r^3 + 6r^2 + 11r + 6 &= 0 \\ (r + 1)(r + 2)(r + 3) &= 0 \end{aligned}$$

Therefore, the characteristic modes are e^{-t} , e^{-2t} , and e^{-3t} . Form the linear combination

$$y_n(t) = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{-3t}$$

2. Create the initial conditions

For a third-order y_n , choose the initial conditions

$$y_n(0) = 0, \quad y'_n(0) = 0, \quad y''_n(0) = 1$$

Differentiating y_n two times then substituting 0:

$$\begin{aligned} y_n(0) &= c_1 + c_2 + c_3 \\ y'_n(0) &= -c_1 - 2c_2 - 3c_3 \\ y''_n(0) &= c_1 + 4c_2 + 9c_3 \end{aligned}$$

Substitute:

$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ -c_1 - 2c_2 - 3c_3 = 0 \\ c_1 + 4c_2 + 9c_3 = 1 \end{cases}$$

Solving,

$$c_1 = \frac{1}{2} \quad c_2 = -1 \quad c_3 = \frac{1}{2}$$

Thus,

$$y_n(t) = \frac{1}{2}e^{-t} - e^{-2t} + \frac{1}{2}e^{-3t}$$

3. Apply $P(D)$

Since $P(D) = D + 2$, calculate

$$\begin{aligned} P(D)y_n(t) &= (Dy_n(t)) + 2y_n(t) \\ &= y'_n(t) + 2y_n(t) \\ &= \frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t} \end{aligned}$$

The causal impulse response is therefore

$$h(t) = [P(D)y_n(t)]u(t) = \frac{1}{2}(e^{-t} - e^{-3t})u(t)$$

Here, the constant coefficient 2 in $P(D) = D + 2$ does **not** produce a $2\delta(t)$ term. The proposed form $2\delta(t) + [P(D)y_n(t)]u(t)$ would add a direct impulse that is not present in this system.

End

Thus the impulse response is

$$h(t) = \frac{1}{2}(e^{-t} - e^{-3t})u(t)$$